Mutual Information in the Cross-Section of Indian Equities: Sign Instability, Feature Overlap, and Conditional-MI Calibration

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Abstract

I screen six price- and volume-based features against forward returns at five horizons in a snapshot-based point-in-time Nifty 500 universe, 2010–2021, using a rank-based mutual information (MI) estimator with a within-date permutation null. All 30 feature–horizon cells have permutation-corrected MI above zero at nominal 95% moving-block bootstrap intervals, and all 30 survive a Holm family-wise adjustment. Only 13 have a mean rank information coefficient (IC) that survives the same adjustment. Most of the gap is not hidden nonlinearity. Much of the measured information comes from monotone relationships that are strong in a given month but change sign across months; an averaged IC cancels these, and sign-blind MI does not. Between features, MI is most informative where rank correlation is weakest: reversal and momentum are each U-shaped in volatility. A first conditional-MI analysis reported synergy for every pair tested; diagnostics traced this to mixing two estimators and a mis-specified noise benchmark, and the corrected analysis shows net redundancy between momentum and volatility and no synergy. MI indicates whether a feature sorts the cross-section of returns, not in which direction.

Read before interpreting results.

- **Sample.** Development period only. Rebalances run from 2010-01 to 2021-12, and every forward window ends on or before 2021-12-31 (a code guard drops any rebalance whose window would cross it; an independent audit re-checked every date used). The later period was not read.
- **Multiple testing.** The screen is a characterization; no feature or horizon was selected from it. In the project's test ledger it is family MI-33 (30 cells), with CMI-34 (four conditional-MI pairs) and MI-35 (two analyses added in the 2026 re-run). These sit after the 32 entries of the original ledger. Nominal and Holm-adjusted results are both reported.
- **Costs.** No portfolios are formed and no returns are claimed; costs do not arise.
- **Point-in-time universe.** Index membership is a snapshot-based approximation built from semi-annual snapshots (first snapshot mid-2010). The 2018-06-30 snapshot is truncated, so membership for Jul–Dec 2018 is approximate.
- **Other.** The panel's date index contains some exchange-closed dates on which most stocks have no data. Months whose rebalance date or forward-window end fell on such a date were dropped (Section 2); a calendar-corrected re-run is planned.

1 Introduction

Information coefficients, the rank correlations between a feature and forward returns, are the standard currency for comparing cross-sectional signals. They detect only monotone relationships

with a consistent sign. Mutual information measures any statistical dependence and is sometimes proposed as a screening tool that catches what correlation misses. This paper asks what MI adds on Indian equity data, and what is required to use it correctly.

I compute a permutation-corrected MI between the cross-sectional ranks of six features (12-1 momentum, one-month reversal, 60-day realized volatility, 21-day log turnover, 21-day overnight-return tilt, and lagged Amihud illiquidity) and forward returns at horizons from one week to one year. The answer has three parts. First, MI flags every feature as informative, but for half the cells the information has no stable direction: monthly rank ICs are often large in absolute value and change sign through time. MI averages the strength of the relationship; IC averages its signed value. Second, MI between features separates monotone overlap, which correlation already describes, from U-shaped relationships, which it does not. Third, conditional MI is easy to misread: a naive implementation produced an apparent synergy for every pair, and the paper documents how that result was diagnosed and corrected.

The contribution is methodological and descriptive. It does not propose a trading rule. Related work by the author studies regime-dependent momentum and reversal ICs in the same market [8]; the sign instability documented here is one reason such conditioning is worth studying.

2 Data and universe

The panel covers NSE-listed stocks from 2010-01-04 to 2026-07-10 (4,237 dates in the panel index, 1,069 tickers), built from Yahoo Finance prices. Historical Nifty 500 membership is a snapshot-based point-in-time approximation from semi-annual index snapshots, expanded into a daily membership flag. Construction, validation and known data defects are documented in the panel repository (see the availability statement).

Choices specific to this paper:

- Only stocks with membership flag equal to 1 on the rebalance date are used. Missing values are dropped, never filled.
- Trailing 21-day features are averaged over each stock's own trading rows. About 31% of (date, stock) cells in the wide panel are empty (closed dates, masked rows, pre-listing), and a calendar window would have left these features empty for most of the universe.
- Amihud illiquidity [1] is computed monthly and attached with a one-month lag: a rebalance in month M uses month $M-1$'s value.

Rebalance dates are the last date in each calendar month present in the panel index, 2010-01 to 2021-12 (144 dates). A date is dropped for a cell if fewer than 30 eligible stocks remain or if its forward window would cross 2021-12-31. At the one-month horizon, cells lose the first five months of 2010 (before the first membership snapshot; momentum, which needs 252 rows of history, loses most of 2010), December 2021 (hold-out guard), and eleven mid-sample months: 2011-08, 2014-03, 2014-06, 2014-07, 2014-08, 2016-10, 2018-02, 2018-03, 2019-09, 2020-11 and 2021-02. Five of these (2011-08, 2014-08, 2016-10, 2018-03 and 2020-11) end on an exchange holiday; the panel's last date in those months appears to be the closed date, carrying data for only a few stocks, so the rebalance cross-section collapses. The cause of the other six has not been verified; a closed date at the end of the forward window would have the same effect. A closed date inside a window also lengthens a row-counted horizon by one trading day. Dropping these months removes about 8% of dates and reduces precision; whether it shifts any estimate will be known after the calendar-corrected re-run. Each cell retains between 97 and 129 dates with 464–490 stocks per date on average.

3 Method

3.1 Permutation-corrected mutual information

On each rebalance date t, the feature and the h-day forward return are converted to cross-sectional ranks scaled to $(0, 1)$. Ranking removes date-level changes in the location and scale of returns, including the common market move, and makes the analysis invariant to monotone transformations; it does not remove differences in stocks' exposures to the market or to other factors. MI between the two rank vectors is estimated with the Kraskov–Stögbauer–Grassberger nearest-neighbour estimator [4] ($k = 3$, scikit-learn). Because this estimator is biased in finite samples, the same quantity is recomputed after 40 random permutations of the return ranks within the date, and

$$\mathrm{MI}_{\mathrm{excess}}(t) = \widehat{\mathrm{MI}}(t) - \tfrac{1}{40} \textstyle\sum_{p=1}^{40} \widehat{\mathrm{MI}}_{\pi_p}(t).$$

The cell estimate is the mean of $\mathrm{MI}_{\mathrm{excess}}(t)$ over dates. The date is the unit of analysis. Pooling stocks from different dates and bootstrapping the pooled sample would place duplicate points in the nearest-neighbour search, which biases this class of estimator upward; estimating one value per date avoids that.

3.2 Inference

Intervals are 95% percentile intervals from a circular moving-block bootstrap [5, 7] over the per-date series (2,000 draws). Monthly rebalances with forward windows longer than a month overlap, so the block length is at least the horizon: 3 months for the one-week, one-month and three-month horizons, 6 for six months and 12 for one year. Because the 30 cells form one family, I also report a Holm family-wise adjustment [3] using z-statistics from the bootstrap standard errors. Mean ICs are computed per date and averaged with equal weight per date; their intervals use the same bootstrap.

3.3 Direction versus rank extremeness

MI is sign-blind, so a positive $\mathrm{MI}_{\mathrm{excess}}$ does not say whether high-feature stocks outperform. For every cell and date I also compute (i) the rank IC, the Spearman correlation of feature and forward return, and (ii) an extremeness MI and extremeness IC, obtained by replacing the forward-return rank u with the rank of $|u - 0.5|$. This measures how far a stock's return rank sits from the cross-sectional median in either direction; it is a rank-based proxy for the size of the move, not the percentage return itself. Before the real-data run I fixed a descriptive classification for each cell: *no signal* if the $\mathrm{MI}_{\mathrm{excess}}$ interval includes zero; *extremeness-led* if extremeness MI is at least half of $\mathrm{MI}_{\mathrm{excess}}$ and its interval excludes zero; *direction-led* if that share is below half and the IC interval excludes zero; otherwise *mixed*. The one-half cut-off is a convention, not a test; the class labels are a taxonomy for reading Figure 1.

To compare MI with rank correlation on one scale, each date's Spearman ρ is converted into the MI of a Gaussian copula with that rank correlation, $-\frac{1}{2}\ln(1 - r^2)$ with $r = 2\sin(\pi\rho/6)$, minus its expected value under independence at that date's cross-section size, $\frac{1}{2}(\pi/3)^2/(n - 1)$. A simulation confirmed that this correction averages zero under independence. The conversion is an approximation, used only for comparisons.

3.4 Conditional mutual information

To ask whether feature A carries information about returns Y beyond feature B, I estimate $I(A; Y \mid B)$ per date with a nearest-neighbour conditional estimator (`npeet`), subtracting a within-date permutation null. Four pairs were fixed before running, all at the one-month horizon: momentum given volatility, volatility given momentum, reversal given momentum, and Amihud

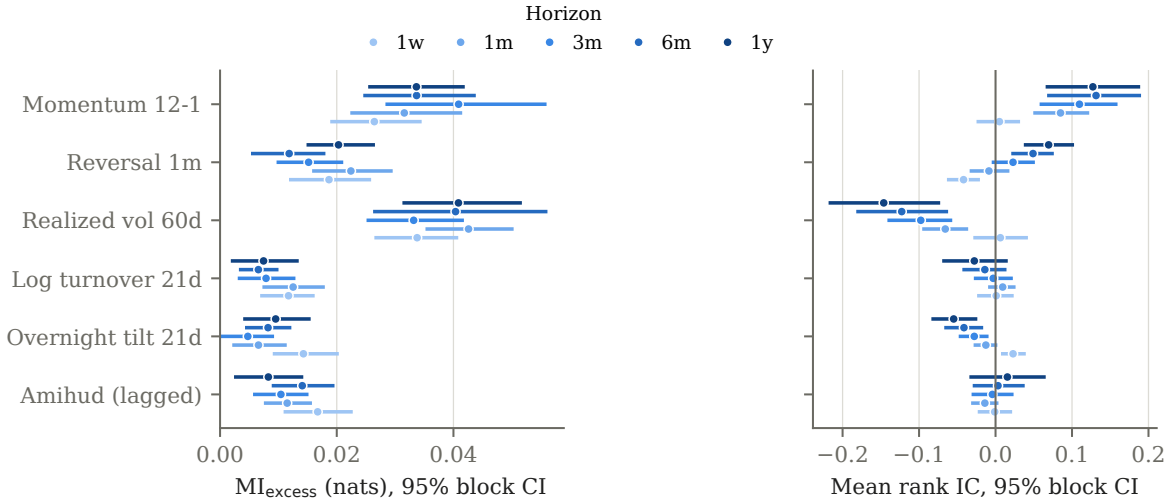


Figure 1: Left: permutation-corrected MI between feature rank and forward-return rank, by horizon. Right: mean monthly rank IC for the same cells. Bars are 95% moving-block bootstrap intervals. All 30 MI intervals exclude zero (30 after Holm adjustment); 15 IC intervals do (13 after Holm adjustment). Development period 2010–2021.

given turnover. Section 4.4 describes how the first version of this analysis failed and how it was corrected.

3.5 Validation gates

No real-data run proceeded until synthetic self-tests passed: a panel with a planted predictive feature and a pure-noise feature (planted $\mathrm{MI}_{\mathrm{excess}} = 0.135$; noise 0.004, 90% interval $[-0.002, 0.011]$); deliberate tests of the membership filter, warm-up gaps, leakage and the hold-out guard; known-redundant and known-additive pairs for the conditional estimator; and, for the 2026 re-run, planted direction-only and volatility-only features for the classification rule.

4 Results

4.1 Every feature is informative; about half have a stable direction

Figure 1 shows the landscape. All 30 $\mathrm{MI}_{\mathrm{excess}}$ intervals exclude zero, and all 30 cells remain significant at the 5% family-wise level after Holm adjustment. Realized volatility (0.033–0.043 nats) and momentum (0.026–0.041) carry the most information, then reversal (0.012–0.022); Amihud, turnover and overnight tilt lie between 0.005 and 0.017. The weakest cell, overnight tilt at three months, is fragile: its lower bound is close to zero with 3- or 6-month blocks and falls just below zero with 12-month blocks. No other cell's conclusion depends on the block length.

The IC panel tells a different story. Momentum has a positive mean IC from one month onward (0.085 at one month, 95% interval $[0.049, 0.122]$; 0.131 at six months) but none at one week.[1] Volatility has a negative IC that strengthens with horizon (-0.066 at one month, -0.146 at one year, $[-0.218, -0.072]$), consistent with the low-volatility pattern documented elsewhere [2]. Reversal is negative at one week (-0.042) and positive at six and twelve months. Turnover's and Amihud's IC intervals include zero at every horizon; overnight tilt's IC is positive at one week and negative from three months, and only the six- and twelve-month values survive Holm

[1]The momentum IC reported by Sahi [9], 0.061, uses 2011–2026 and a different return harness. Restricting this paper's one-month series to 2011–2021 gives 0.084, so the different start date does not explain the gap; the inclusion of 2022–2026 and harness differences remain possible causes and have not been separated.

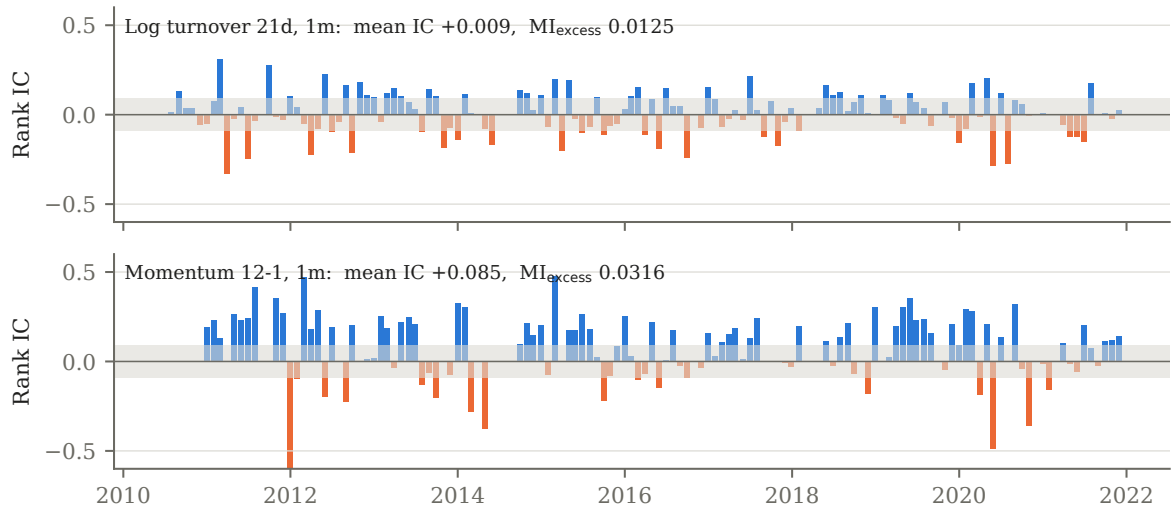


Figure 2: Monthly rank IC at the one-month horizon. The grey band is $\pm 1.96/\sqrt{n-1}$, a reference for sampling noise if stocks were independent; it is not a calibrated null, since returns are cross-sectionally correlated. Turnover's monthly ICs fall outside the band in both directions and average to almost zero; momentum's are mostly positive.

Table 1: One-month horizon. Share of months with IC above / below the reference band; mean IC; and Gaussian-copula MI implied by the mean IC versus by each month's own IC (then averaged). Negative null-corrected values are shown as ≈ 0.

	Months outside band			Implied MI (nats)		
Feature	above	below	Mean IC	from mean IC	from monthly IC	MI$_{excess}$
Momentum 12-1	57%	16%	0.085	0.0028	0.0242	0.0316
Reversal 1m	25%	25%	−0.009	≈ 0	0.0121	0.0224
Realized vol 60d	19%	46%	−0.066	0.0012	0.0201	0.0426
Log turnover 21d	28%	19%	0.009	≈ 0	0.0067	0.0125
Overnight tilt 21d	13%	21%	−0.013	≈ 0	0.0034	0.0066
Amihud (lagged)	18%	28%	−0.014	≈ 0	0.0067	0.0115

adjustment. In total, 15 IC intervals exclude zero nominally and 13 survive Holm adjustment. Under the descriptive classification, 13 cells are direction-led, 4 extremeness-led, 13 mixed and none without signal. Volatility carries both kinds of information: alongside its negative IC, its extremeness IC is 0.16–0.17 at every horizon. High-volatility stocks end further from the median, in either direction.

4.2 Much of the information has no stable sign

Why do so many cells carry information without a stable direction? Figure 2 shows the pattern, and Table 1 quantifies it. At one month, turnover's IC lies above the reference band in 28% of months and below it in 19%, and the two cancel to a mean of 0.009. Momentum is the contrast: 57% above, 16% below. Across all 30 cells, between 28% and 78% of months fall outside the band, against 5% expected if stocks were independent and there were no relationship. Some of this excess reflects cross-sectional correlation among stocks, which widens the true sampling distribution of a monthly IC. A likely economic reading is that several features are correlated with exposures to common factors or sectors whose returns change sign from month to month; the month's association is real, but its direction follows the factor return. This mechanism is not tested here.

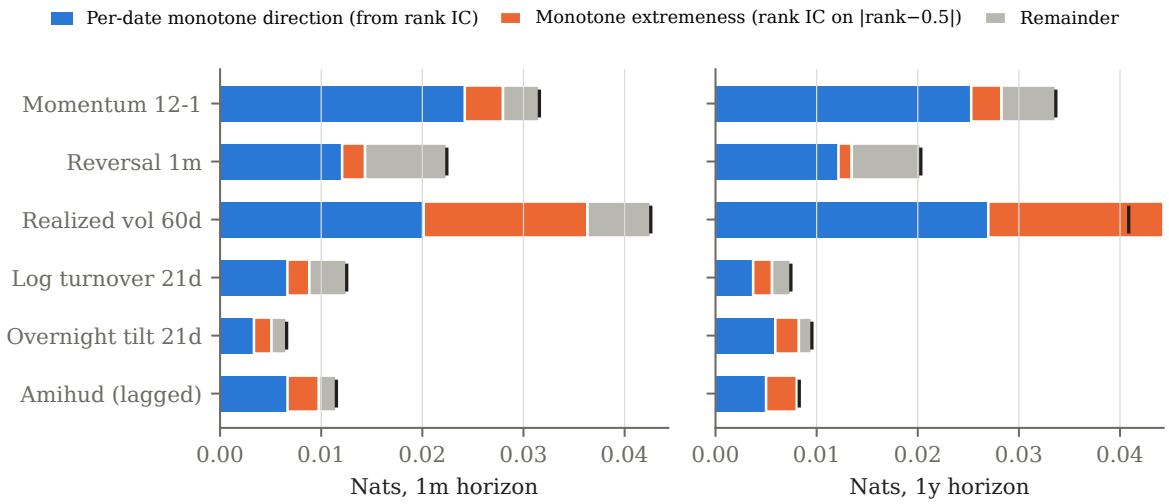


Figure 3: Approximate accounting for $\mathrm{MI}_{\mathrm{excess}}$ (black tick) at one month and one year: MI implied by each month's rank IC (blue), by each month's extremeness rank IC (orange), and the remainder (grey). Components use a Gaussian-copula conversion and are not strictly additive.

The last three columns put this on the MI scale without using the MI estimator. The information implied by each month's own IC is about nine times what the mean IC implies for momentum, whose direction is the most stable, and 17 times for volatility; for the other four features the mean IC implies essentially none. As an approximate accounting (Figure 3), month-by-month monotone direction corresponds to a median of 60% of $\mathrm{MI}_{\mathrm{excess}}$ across the 30 cells (range 35–91%), and adding monotone extremeness brings the median to 83% (interquartile range 71–88%). These shares are not an exact decomposition: the components are approximate and not strictly additive. They may also overstate the monotone part if the MI estimator understates weak dependence (Section 5). The sign-frequency evidence in Table 1 does not depend on the MI estimator.

For signal research, a high MI score answers "does this feature sort stocks by future return in a typical month?" and not "in which direction, on average?". Ranking features on MI alone would place turnover and Amihud, whose average direction is indistinguishable from zero, alongside features with a measurable average direction.

4.3 Feature overlap: monotone and U-shaped

The same comparison between features (Figure 4) separates two kinds of overlap. Turnover and Amihud illiquidity are almost perfectly rank-correlated ($\rho = -0.950$); this is mechanical, since turnover is the denominator of Amihud's measure. Their MI (1.27 nats) is within 4% of what that correlation implies, so rank correlation describes this overlap completely. (The original screen had blanked this cell automatically because it exceeded a sanity ceiling; a follow-up check found no tied values and confirmed the dependence was genuine.) By contrast, reversal and volatility are uncorrelated on average ($\rho = -0.006$, 95% interval $[-0.050, 0.034]$) yet share 0.090 nats, 2.8 times the Gaussian-implied amount. A stock that moved sharply in either direction last month tends to be a high-volatility stock: a U-shaped relationship that correlation averages away. Momentum and volatility ($\rho = -0.105$, MI 0.110, 2.0 times implied) show the same pattern: extreme winners and extreme losers both tend to be volatile. I have not checked these shapes with binned conditional means, so "U-shaped" is the natural reading of the MI–correlation gap rather than a fitted result. This is where MI earns its place next to correlation: in revealing non-monotone structure between features, rather than in the case that first motivated the analysis.

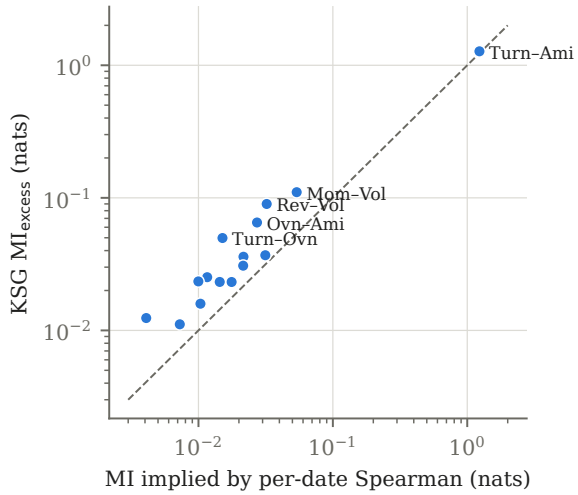


Figure 4: MI between pairs of features (vertical) against the MI implied by their monthly Spearman correlation (horizontal), log scales. Points on the dashed line are relationships rank correlation fully describes. Labelled points lie furthest above it. Mom = momentum, Rev = reversal, Vol = volatility, Turn = turnover, Ovn = overnight tilt, Ami = Amihud.

4.4 Conditional MI: an apparent synergy and its correction

The first conditional run expressed each result as a survival ratio: conditional MI of A given B divided by A's marginal MI. All four pre-registered pairs returned ratios above 100% (117%, 117%, 128% and 169%), which would mean every feature carries more information once its partner is known. A uniform result across unrelated pairs is more likely an artifact than a market property, so three estimator checks were run before interpretation:

- **Two estimators, two scales.** The numerator used the `npeet` estimator and the denominator scikit-learn's. On identical inputs their marginal MI estimates differed by a factor of 0.68–1.42 depending on the feature.
- **Conditioning on noise is not neutral.** Conditioning a feature on a pure random variable should leave its information unchanged. Instead the "surviving" fraction ranged from 44% to 113% across features, so the right benchmark is this feature-specific value, not 100%.
- **Conditioning on itself does collapse.** Conditioning a feature on itself left between −8% and +6%, so the estimator can detect redundancy.

The corrected analysis uses one estimator throughout. For each date it computes A's permutation-corrected conditional information given B, and the same quantity with B replaced by random noise (averaged over 10 independent noise draws). With the same estimator and inputs, conditioning on independent noise approximates A's marginal information $I(A;Y)$. The per-date difference therefore estimates

$$I(A;Y \mid B) - I(A;Y),$$

the interaction information of A, B and Y [6]. Negative values indicate net redundancy (part of what A says about Y is already in B); positive values indicate net synergy. The measure is net: it cannot separate redundancy from synergy when both are present, which would require a partial-information decomposition. Intervals bootstrap the per-date differences, so sampling variation in both terms, including the benchmark, is carried into the interval. Before use, the method was checked on synthetic data with a planted redundant pair (difference −0.106 nats against a benchmark of 0.113) and a planted additive pair (+0.017, interval excluding zero).

Table 2 shows no synergy for any pair. Momentum and volatility show net redundancy in both directions: as a ratio of point estimates, conditioning on volatility lowers momentum's estimated information by about a third (0.018 of 0.053 nats), and conditioning on momentum lowers volatility's by about 29%. No interval was computed for these ratios, so they are descriptive. Each feature retains most of its information, so this is partial overlap, not duplication. Reversal given momentum and Amihud given turnover are inconclusive: their intervals include zero, which

Table 2: Corrected conditional MI at the one-month horizon (nats). Benchmark: A's conditional information given random noise. Difference: given B minus benchmark, an estimate of interaction information. Re-run 2026-10-13; the original 2026-09 difference is shown for comparison.

$A \mid B$	Dates	Given B	Benchmark	Difference	95% interval	Original
Momentum | volatility	108	0.0349	0.0527	−0.0177	[−0.0285, −0.0078]	−0.0176
Volatility | momentum	108	0.0495	0.0697	−0.0202	[−0.0305, −0.0096]	−0.0199
Reversal | momentum	108	0.0305	0.0335	−0.0030	[−0.0113, 0.0052]	−0.0025
Amihud | turnover	127	0.0188	0.0206	−0.0018	[−0.0107, 0.0068]	−0.0013

means neither redundancy nor an independent increment is established. The Amihud result is not at odds with the near-perfect correlation of the two features. Amihud's own information about returns is small (0.023 nats), so there is little to remove, and the interval is wide relative to it.

5 Diagnostics

Exact reproduction. The 2026 re-run recomputed the screen from the same panel and code with the original random seed. It reproduced all 30 point estimates and date counts to floating-point precision (largest difference 10^{-16}). The conditional-MI differences reproduced within 0.0005 nats under newer library versions. Reviewing the code during this exercise showed that the original driver set 15 permutations and 500 bootstrap draws after the functions had already fixed their defaults, so the screen actually ran with 40 and 300; the effective values are reported here.

Interval method and multiplicity (revision diagnostics). Moving-block intervals were 1–2% wider than iid intervals at one week and one month, 10% wider at three and six months and 14% wider at one year. With block lengths of 3, 6 and 12 months for every cell, only overnight tilt at three months changes conclusion (Section 4.1). The Holm adjustments in Section 4 were computed after the re-run from the archived per-date series.

Estimator calibration. Table 3 compares the permutation-corrected KSG estimate with the known MI of bivariate Gaussian data at $n = 480$, the typical cross-section size. At moderate and strong dependence the estimate is within 5% of the truth, except at $\rho = 0.3$, where it is 17% high. At weak dependence ($\rho \leq 0.2$, the range of most feature–return cells) the calibration is imprecise: across 30 replications the estimate averaged 38–40% of the true value, with a standard error of about 0.004 nats. That shortfall is within noise at $\rho = 0.1$ and about three standard errors at $\rho = 0.2$. Reported $\mathrm{MI}_{\mathrm{excess}}$ values for weak cells may therefore understate the true MI.

Table 3: KSG calibration on bivariate Gaussian data, $n = 480$, 30 replications, 10 permutations each.

ρ	0.1	0.2	0.3	0.6	0.9	0.98
True MI (nats)	0.0050	0.0204	0.0472	0.2231	0.8304	1.6145
Estimated $\mathrm{MI}_{\mathrm{excess}}$	0.0020	0.0078	0.0550	0.2122	0.8472	1.6085
SD across replications	0.0195	0.0234	0.0307	0.0450	0.0616	0.0672

Reproducibility. Table 4 lists the settings needed to reproduce every number in the paper; a single notebook in the repository regenerates them.

Table 4: Run settings.

Sample	Month-end rebalances 2010-01 to 2021-12 (144); 97–129 usable dates per cell; 464–490 stocks per date on average
MI estimator	KSG, $k = 3$, scikit-learn 1.6.1; ranks scaled to $(0, 1)$; 40 within-date permutations; seed 20260809
Conditional MI	npeet-plus 0.2.0, $k = 3$; 40 permutations for the conditional term; benchmark from 10 noise draws with 7 permutations each
Intervals	Circular moving-block bootstrap, 2,000 draws, blocks 3/3/3/6/12 months for 1w/1m/3m/6m/1y; seed 20260810 (original run: iid, 300 draws, 90%)
Software	Python 3.13.16, NumPy 2.1.3, pandas 2.2.3 (re-run, 2026-10-13)

6 Limitations and what this paper does not establish

- **No trading claim.** The paper characterizes information. It forms no portfolios, models no costs, and does not show that any feature is profitable.
- **No mechanism.** The sign instability is documented, not explained; the factor-exposure reading in Section 4.2 is untested. The U-shapes between features are inferred from the MI–correlation gap, not fitted.
- **Approximate accounting.** The Gaussian-copula conversion is an approximation and the direction / extremeness / remainder shares are not an exact decomposition.
- **Net conditional measure, one horizon.** Interaction information is net of redundancy and synergy. The four pairs were tested only at one month; conditional estimation in three dimensions is noisier, and longer horizons were not pre-specified.
- **Weak-dependence calibration.** Absolute MI values below about 0.02 nats should be read as approximate.
- **Calendar.** About 8% of rebalance months were lost to closed or incomplete dates in the panel index (Section 2); results will be re-estimated on a corrected calendar.
- **Universe approximation.** Membership comes from semi-annual snapshots, including the truncated 2018-06-30 snapshot.

7 Conclusion

In Indian large- and mid-cap equities, permutation-corrected mutual information indicates that all six common price and volume features sort the cross-section of future returns, and this holds after a family-wise adjustment. Much of that information, however, reflects relationships that are strong in a given month and unstable in sign across months. Rank IC, by averaging signed values, answers the question a portfolio needs answered; MI answers whether there is anything to condition or time. Used between features, MI was most informative where correlation was weakest, revealing that reversal and momentum are each related to volatility in a U-shape. And conditional MI needed care: a single estimator and a feature-specific noise benchmark turned an apparent uniform synergy into partial, net redundancy between momentum and volatility. The procedural lesson is that a surprisingly uniform result is a reason to test the estimator before interpreting the market.

Use of AI tools. Code for this project was written with the assistance of an AI coding assistant (Anthropic's Claude), working to specifications set by the author, and the manuscript was drafted with the same assistance. The author designed the research questions and decision rules and ran the main analyses. Some diagnostics added during revision were computed by the assistant from archived run outputs; these are labelled as revision diagnostics in the text. The author reviewed all code and text and takes full responsibility for the content.

Data and code availability. Code: https://github.com/mriganksahi-beep/indian-equity-research/tree/main/p3_mi_screen. The point-in-time panel construction is documented in https://github.com/mriganksahi-beep/indian-equity-research/tree/main/data_pipeline. Price data from Yahoo Finance and index-membership snapshots from a licensed vendor are not redistributed; the repository contains the scripts to rebuild the panel.

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